

THE LEDGER OF SHADOWS

Technical Investigation Report

1. Motivation Behind the Approach

Financial fraud detection is a highly sensitive classification problem where the cost of missing a fraudulent transaction is significantly higher than incorrectly flagging a correct one. The objective of this investigation was not only to maximize prediction accuracy but also to minimize total investigation loss.

A fraud detection model was developed to:

- assign a fraud risk probability score to each transaction
- identify suspicious transaction patterns
- detect fraudulent transactions
- optimization of thresholds based on investigation cost

Since fraud datasets highly imbalanced, special attention was given to imbalance handling and threshold optimization.

2. Dataset Observations

The dataset contained anonymized transaction level financial records with a binary target variable (Class) indicating fraud or legal transactions.

Key Observations

- No missing values were present.
- Fraudulent transactions formed a very small minority of the dataset
- Transaction amounts were highly accumulated with several outliers
- Features $V_1, V_2, \dots, V_{28}$ were already PCA transformed which reduced feature engineering requirements.

Exploratory Data Analysis (EDA) showed strong class imbalance and clear behavioural differences between fraudulent and legal transactions.

3. Class Imbalance Analysis

The dataset was extremely imbalanced, with legal transactions dominating the dataset. Therefore, accuracy alone was not a reliable evaluation method because a model predicting all transactions as legal could still achieve very high accuracy.

To properly evaluate performance, the following metrics were prioritized
Precision, Recall, F1 Score, ROC-AUC, PR-AUC, Investigation Loss

The total loss function used was $L = FP + 25FN$ where false negatives were heavily penalized because undetected fraud is dangerous.

4. Preprocessing Decisions

- Data Cleaning

No null values were found, so no imputation was required.

- Feature Scaling

Amount and Time features were standardized using StandardScaler to improve model performance and prevent large value features from dominating training.

- Dataset Splitting

The dataset was divided into:

- training set
- validation set
- and test set

The validation set was specifically used for threshold tuning.

5. Models Used

- Logistic Regression

Used as a simple and interpretable baseline model.

`class_weight='balanced'` was applied to reduce majority-class bias.

It's a good baseline model, fast and easily interpretable but has limited capability to capture non-linear fraud.

- **Random Forest Classifier**

Used because ensemble methods perform well on structured tabular datasets and can capture complex feature interactions. This model has strong non-linear learning abilities thus also reducing overfitting.

This model achieved the best operational performance.

- **XGBoost Classifier**

Used due to its strong predictive performance on structured datasets.

It has high ROC-AUC and effective gradient boosting but it had produced excessive false positives.

6. Imbalance Handling Method

To address class imbalance, **SMOTE (Synthetic Minority Oversampling Technique)** was applied on the training dataset.

SMOTE was applied to:

- generates synthetic fraud samples (minority class)
- balances class distribution
- improves minority class learning
- helps models better detect fraud patterns

SMOTE was applied only on training data to avoid data leakage into validation and testing.

7. Threshold Selection Logic

Thresholds from 0.01–0.99 were tested instead of using the default 0.5 cutoff. The threshold minimizing investigation loss, defined as $L = FP + 25FN$, was selected. This prioritized reducing costly false negatives while tolerating some false positives.

Separate tuning was performed for Logistic Regression, Random Forest, and XGBoost using the validation set.

8. Final Model Performance

- **Logistic Regression**

Total Investigation Loss: 331

Analysis

The model achieved strong recall but moderate precision. It successfully detected most fraudulent transactions but generated a noticeable number of false alarms.

- **Random Forest Classifier**

Total Investigation Loss: 297

Analysis

This model achieved the best overall balance between fraud detection and operational cost. It produced highest precision, highest F1-score, lowest investigation loss. It was selected as the final recommended model.

- **XGBoost Classifier**

Total Investigation Loss: 383

Analysis

Although XGBoost achieved the highest ROC-AUC score, it generated too many false positives, making it operationally inefficient under the defined cost function.

9. Interpretability Findings & Suspicious Patterns

Key Findings

- Fraudulent transactions formed distinguishable probability clusters.
- Unusual transaction amounts contributed significantly to fraud prediction.
- Ensemble models captured hidden nonlinear fraud patterns more effectively than Logistic Regression.

10. Limitations

- Anonymized features reduced real-world interpretability.
- SMOTE-generated synthetic samples may not perfectly represent real fraud patterns.
- False positives still require human investigation.

11. Final Recommendation to MI6

Based on all evaluation metrics and operational cost analysis, the Random Forest Classifier is recommended.

Reasons:

- lowest total investigation loss,
- strong fraud detection capability
- balanced precision and recall
- fairly good performance on highly imbalanced data
- better operational efficiency than other models.

Recommended deployment strategy:

1. Use Random Forest as the primary fraud detection engine.
2. Continuously monitor investigation loss and fraud trends.
3. Periodically retrain the model using newer transaction data.

The developed system demonstrates that machine learning can effectively reduce financial fraud risk while maintaining manageable investigation costs for large-scale transaction monitoring systems.